Southern New Hampshire University

CS 255 Business Requirements Document

Complete this template by replacing the bracketed text with the relevant information.

This template lays out all the different sections that you need to complete for Project One. Each section has guiding questions to prompt your thinking. These questions are meant to guide your initial responses to each area. You are encouraged to go beyond these questions using what you have learned in your readings. You will need to continually reference the interview transcript as you work to make sure that you are addressing your client's needs. There is no required length for the final document. Instead, the goal is to complete each section based on your client's needs.

Tip: You should respond in a bulleted list for each section. This will make your thoughts easier to reference when you move into the design phase for Project Two. One starter bullet has been provided for you in each section, but you will need to add more.

System Components and Design

Purpose

What is the purpose of this project? Who is the client and what do they want their system to be able to do?

- Create a system to train students for their driver's test.
- (Owner)Liam and (IT Offers) Ian are the clients from driverpass.
- Want them to take online classes, practice tests and provide an optional on-the-road training.
- Deliver administrative, scheduling, and tracking tools so DriverPass staff can run daily operations, manage instructors, and track vehicle resources efficiently.

System Background

What does DriverPass want the system to do? What is the problem they want to fix? What are the different components needed for this system?

- Allow staff to access data online from any computer or mobile device, with the ability to download reports to work offline in Excel without causing data duplication. Identified high failure rates on DMV driver's tests due to inadequate driver preparation and limited access to structured training programs.
- DriverPass aims to bridge this gap by offering a hybrid model featuring structured online coursework, simulated practice exams, and direct behind-the-wheel instructions.
- • **Student Area:** A website where students sign up, take online lessons, take practice tests, view their past scores, and book or change their driving appointments.

- **Appointment & Car Scheduler:** A calendar tool that books two hour driving lessons by automatically matching a student with an available instructor and one of the company's 10 cars.
- **Staff & Account Controls:** A setup that gives each worker the right level of account access (owner, IT, secretary, or instructor) and lets students reset forgotten passwords on their own.

- **Activity History & Tracking:** An automatic log that tracks who booked, changed, or canceled an appointment, and when they did it, so the office always knows who made updates.

- **Instructor Page:** A simple screen where driving teachers log the start and end times of each lesson and leave notes on how the student drove.
- **DMV Updates Feed:** An alert feature that notifies the team whenever the local DMV updates its driving laws, rules, or test questions.

Objectives and Goals

What should this system be able to do when it is completed? What measurable tasks need to be included in the system design to achieve this?

Account Types

- Create separate accounts for the boss, IT guy, secretary, driving instructors, and students so each person only sees what they need.
- Let students reset their own passwords online if they forget them.

Sign-Up and Info

- Make simple forms to take student names, phone numbers, addresses, and credit card payments.
- Make sure the student gets picked up and dropped off at the same spot.

Practice Tests & DMV Updates

- Give students a screen that shows their test scores, how long tests took, and if they passed or failed.
- Notify the staff whenever the DMV changes any laws or test questions.

Scheduling Cars & Drivers

- Let students book, change, or cancel 2-hour driving lessons online, or let the secretary do it over the phone.
- Match each lesson with one of the 10 cars and an available driver without double-booking.

Driver Feedback

- Give instructors a quick page to write down lesson start/end times and notes on how the student did.

Packages

- Set up the 3 training packages (6 hours in-car, 8 hours with classroom, or 12 hours with classroom and online tests).
- Let the boss turn a package on or off with a simple switch without calling a programmer.

Tracking & Reports

- Keep a log of who made, changed, or canceled any appointment so the office always knows what happened.
- Let the boss download company info straight into an Excel file to view at home.
- Keep everything online in the cloud so backups happen automatically.

Requirements

Nonfunctional Requirements

In this section, you will detail the different nonfunctional requirements for the DriverPass system. You will need to think about the different things that the system needs to function properly.

Performance Requirements

What environments (web-based, application, etc.) does this system need to run in? How fast should the system run? How often should the system be updated?

- The DriverPass system will run online in the cloud, so people can use it from both computers and mobile devices. All information like schedules, training materials, and account details will be accessed and updated through the online system. The system should remain fast and responsive during normal use, providing smooth access to scheduling, training materials, and account information without any delays.

To keep the system reliable, the cloud service should have very little downtime and handle backups automatically. This means DriverPass staff won't need to worry about managing storage or security themselves. The system will get regular monthly updates, plus any extra updates needed when the DMV

changes its rules or releases new practice test content. All changes to data must happen online; the only thing users can do offline is view reports they've already downloaded.

Platform Constraints

What platforms (Windows, Unix, etc.) should the system run on? Does the back end require any tools, such as a database, to support this application?

- The DriverPass system will run in the cloud as a web application, so it does not depend on a specific operating system like Windows. Any device with an internet connection and a web browser can use the system. On the back end, the system will need a cloud hosted database to store customer information, appointments, packages, and driver notes. The cloud platform will also handle backups and security so DriverPass staff do not need to manage those tools themselves.

Accuracy and Precision

How will you distinguish between different users? Is the input case-sensitive? When should the system inform the admin of a problem?

- The system will distinguish between users by assigning each person a unique account with a username and password. Input such as names or search terms will not be case-sensitive, so users don't have to worry about typing in capital letters. The system should notify the admin whenever there is a problem, such as failed logins, scheduling conflicts, or issues saving customer information, so the admin can fix the issue quickly.

Adaptability

Can you make changes to the user (add/remove/modify) without changing code? How will the system adapt to platform updates? What type of access does the IT admin need?

- The system should allow the admin to add, remove, or update user accounts without needing to change any code. All user management should be handled through the system's interface. Because the system runs in the cloud, it can adapt to platform updates automatically, such as browser updates or mobile device changes. The IT admin should have full access to manage

Security

What is required for the user to log in? How can you secure the connection or the data exchange between the client and the server? What should happen to the account if there is a "brute force" hacking attempt? What happens if the user forgets their password?

- *Users must log in with a username and password. The connection should be protected so their information stays private. If someone keeps trying to break in by guessing passwords, the account should lock for safety. If a user forgets their password, the admin should be able to reset it so they can get back in.*

Functional Requirements

Using the information from the scenario, think about the different functions the system needs to provide. Each of your bullets should start with "The system shall . . ." For example, one functional requirement might be, "The system shall validate user credentials when logging in."

- The system shall allow customers to purchase training packages.
- The system shall validate user credentials when logging in.
- The system shall allow the admin to reset user passwords.
- The system shall allow instructors to update student driving notes.

User Interface

What are the needs of the interface? Who are the different users for this interface? What will each user need to be able to do through the interface? How will the user interact with the interface (mobile, browser, etc.)?

- The DriverPass system will be used by customers, instructors, and the admin. Customers need to log in, buy packages, schedule lessons, and check progress. Instructors need to see their schedule and update student notes. The admin needs to manage accounts and reset passwords. Everyone will use the system through a web page that works on computers and phones.

Assumptions

What things were not specifically addressed in your design above? What assumptions are you making in your design about the users or the technology they have?

- Some details were not fully covered, such as payment processing, exact database setup. I am assuming users have internet access, a device with a web browser, and basic computer skills.

Limitations

Any system you build will naturally have limitations. What limitations do you see in your system design? What limitations do you have as far as resources, time, budget, or technology?

- The system depends on internet access and won't work offline. It also relies on cloud services, which may have small outages. Time, budget, and staff limits could slow down new features or improvements. Some advanced tools may not be included at first because of these limits.

Gantt Chart

Please include a screenshot of the GANTT chart that you created with Lucidchart. Be sure to check that it meets the plan described by the characters in the interview.

